

Vaibhav Kumar

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PROFESSIONAL SUMMARY

M.Sc. Computer Science graduate (2026) specializing in **Machine Learning Systems, Deep Learning (MoE), and Agentic AI Pipelines**. Dual **Oracle Cloud Certified** in Generative AI & Data Science. Experienced in building sub-15ms REST APIs (FastAPI), multi-agent LLM systems with tool calling, and large-scale dataset benchmarking. Based in Hyderabad; open to on-site and remote roles across India.

EDUCATION

Central University of Rajasthan 2021 – 2026
Integrated M.Sc. in Computer Science (Machine Learning & Systems Focus) Ajmer, Rajasthan

TECHNICAL SKILLS

Machine Learning & Deep Learning: PyTorch, TensorFlow, Scikit-learn, LightGBM, XGBoost, MoE Architectures, OpenCV
Generative AI & Agents: LangChain, Groq API (Llama 3 70B), Vector DBs (FAISS, ChromaDB), RAG Pipelines, Prompt Engineering
Backend & Data Systems: Python (FastAPI, Flask), SQL (PostgreSQL, MySQL), Pandas, NumPy, Cleanlab (Data-Centric ML)
Cloud, DevOps & Tools: Oracle Cloud Infrastructure (Dual OCI Certified), Docker, Git, Linux, Streamlit, Vercel

WORK EXPERIENCE

- Research Intern – Quantum ML & Systems Modeling** Jan 2026 – May 2026
University of Hyderabad (SCIS) Hyderabad, India
- Benchmarked machine learning architectures across real-world datasets (68,000+ records) under faculty supervision.
 - Implemented reproducible validation routines, anomaly detection filters, and statistical evaluation metrics for dissertation research.
- Data Science Intern** Jun 2025 – Aug 2025
Celebal Technologies Remote, India
- Engineered automated data preprocessing pipelines and SQL transformations across 100,000+ transactional records.
 - Implemented IQR outlier filters, StandardScaler normalization, and automated exploratory analysis dashboards via Pandas and Seaborn.
- AI & Machine Learning Intern** Jun 2024 – Dec 2024
YBI Foundation Remote, India
- Constructed multi-model classification pipelines using Scikit-learn with automated GridSearchCV hyperparameter optimization.
 - Developed interactive Streamlit web applications and prototype RAG Q&A systems using FAISS vector indexing.

KEY TECHNICAL PROJECTS

- Agentic Honeypot for Scam Detection & Intelligence Extraction**   **Live on Vercel**
Tech Stack: Python, FastAPI, LangChain, Groq API (Llama 3 70B), Librosa, Pydantic, Vercel Serverless
- Built and deployed multi-agent cybersecurity honeypot autonomously classifying phishing attacks and extracting threat vectors (UPI, links).
 - Implemented multi-turn adaptive conversational state machine and integrated Librosa acoustic feature analysis for voice deepfake detection.
- DaAIR-mini: Lightweight Image Restoration via Mixture-of-Experts**  **PyTorch / MoE**
Tech Stack: Python, PyTorch, Mixture-of-Experts (MoE), Soft-Attention Routing, Computer Vision
- Designed 50,343-parameter MoE architecture achieving 27.95 dB PSNR / 0.7923 SSIM with 100× parameter reduction vs. Restormer baselines.
 - Implemented low-rank DaLe factorised expert networks with soft-attention gating and EMA parameter stabilization during training.
- Metadata Threat Radar: Real-Time Ransomware Detection API**  **Sub-15ms REST API**
Tech Stack: Python, FastAPI, Scikit-learn, RandomForest, Pydantic V2, NumPy, Uvicorn
- Engineered real-time ransomware attack detection REST API reaching 91% attack recall and sub-15ms inference latency.
 - Extracted Shannon entropy and file system operation velocity from 8.37MB telemetry logs with Pydantic schema validation.

CERTIFICATIONS

- Oracle Cloud Infrastructure 2025 Generative AI Certified Professional** (Oracle Cloud, Valid to 2027)
- Oracle Cloud Infrastructure 2025 Data Science Professional** (Oracle Cloud, Valid to 2027)
- McKinsey Forward Program – Problem Solving & Business Analytics** (McKinsey & Company, 2024)